

WashU

Mobility Project

2025 Digital Transformation Summer Corps

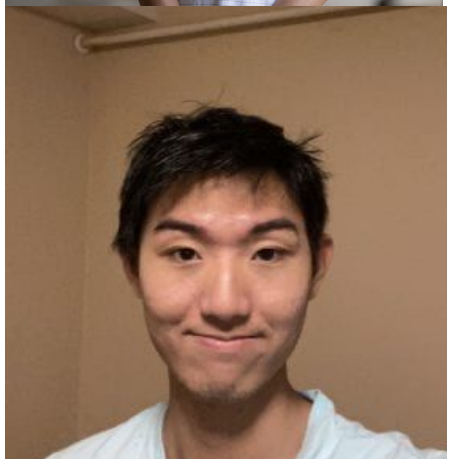
Justin Park, Richie Jiang, Jonathan Lin

Collaborators

- Faculty leads
  - Joshua Jackson, William Yeoh
- Staff leads
  - Alex Chen, Dan Maranan, Samuel Liu
- Schools
  - Arts & Sciences, McKelvey School of Engineering
- Student contributors
  - Justin Park, Richie Jiang, Jonathan Lin







WashU

The Problem/Question

Research Question:
How do human mobility patterns reflect behavioral outcomes?

Examples:
- Where do students go on Friday nights?
- How do mobility patterns correlate with GPA?

Our Goal: Enable researchers to explore these questions through data processing and visualization.

Why This Matters?
Understanding mobility reveals insights about social integration, academic success, and mental health

The Problem/Question

The Challenge

- **Fragmented Data**
  - Mobility data exists in incompatible formats across sources (WashU & Dartmouth)
  - GPS, WiFi, Bluetooth use different schemas and coordinates
  - No unified analysis across datasets
- **Research Barriers**
  - Researchers can’t easily access fragmented data
  - Cross-institutional students nearly impossible

Overview

What We Built

ETL Pipeline (Data aggregation + processing):

- Converts multi-format data sources into centralized geoparquet format
- User information hashed for privacy protection
- Optimized for fast loading, query performance, and space efficiency

Interactive Web Platform (Data Visualization):

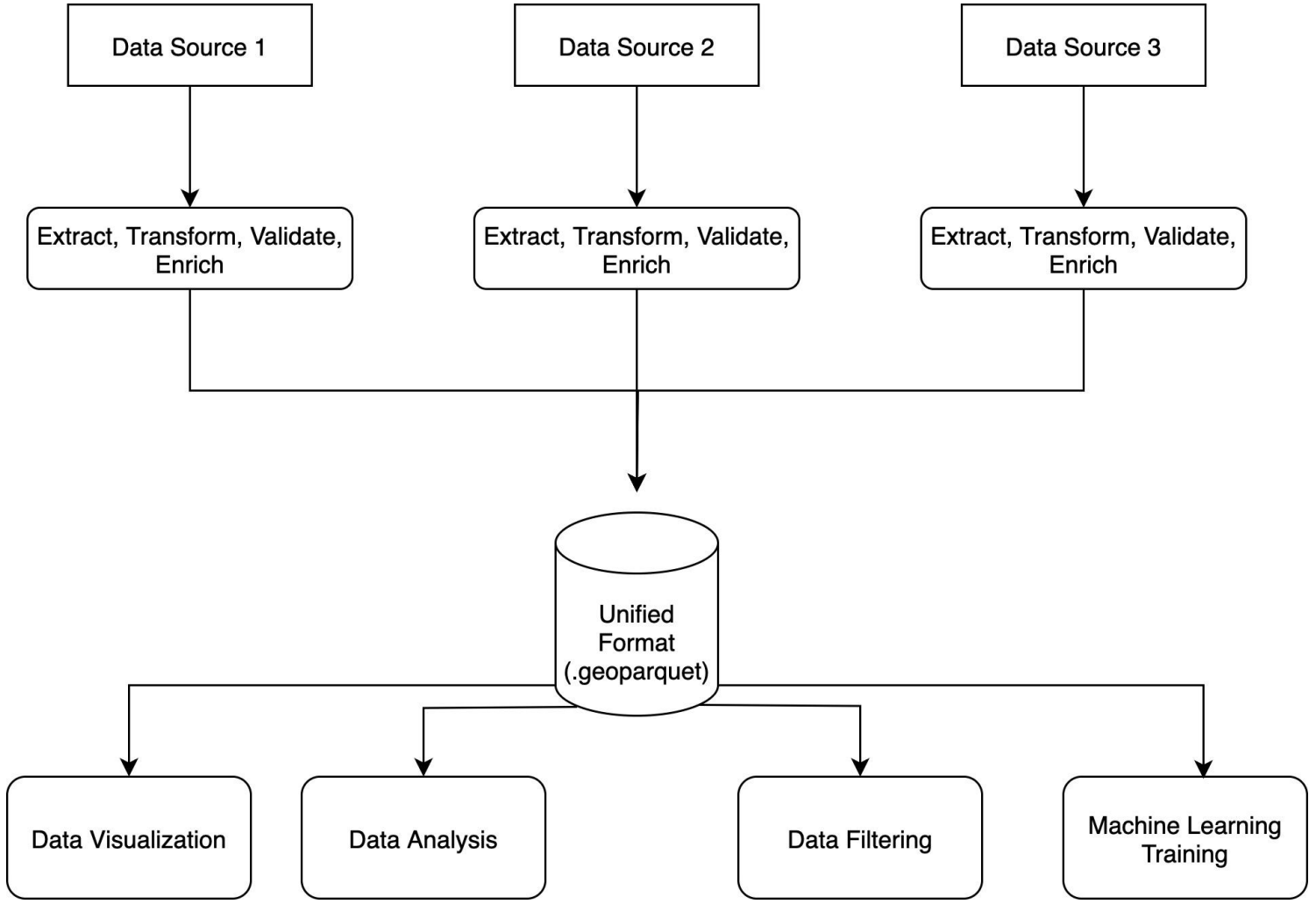
- **Bounding Box Querying:** Get data within specific geographic regions
- **Timelapse Visualization:** View temporal patterns and individual trajectories
- **Smart Routing:** Connect points with real world paths.
- **Advanced Filtering:** Time, gender, school year, and custom parameters
- **Filtered data export:** Export filtered data for deeper analysis.

Approach

- **Phase 1: Data Infrastructure + visualization experiments**
  - Built ETL pipeline for multi-source data standardization
  - Experimented with different visualization tools

- **Phase 2: Platform Development**
  - Developed interactive web interface (both frontend and backend)
  - Integrated mapping and routing capabilities
  - Added filtering and visualization features
  - Performance optimization

ETL Pipeline



Web Demo

Broader Impact

Vision:

Build a WashU mobility infrastructure that enables interdisciplinary collaboration around human movement patterns.

Who Benefits:

WashU Researchers Across Disciplines:

- Psychology: Study behavioral patterns and human psychology.
- Political Science: Analyze social movement and gathering patterns.
- Urban Health: Research campus wellness and activity behaviors

Scalability & Reuse:

- ETL pipeline adaptable to other mobility datasets
- Web platform reusable for geospatial data visualization

Next Steps

- **Are there follow-up plans for this project?**
  - Append more visualization options to the tools
  - Modify the tool so it is more user-friendly and easy to navigate
  - Address bugs and potential technical issues
  - Start working on data analysis features

Thank you